

LA GRANDE 4, QC, Canada

WMO: 710030

Lat: 53.755N

Lon: 73.675W

Elev: 1005

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-30.7	-43.2	0.5	-35.5	-38.5	0.6	-30.6	24.6	3.5	21.7	3.4	1.8	60	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	79.5	61.8	75.8	60.2	72.2	59.1	65.6	73.5	63.5	70.9	61.7	68.5	10.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	89.0	68.2	60.9	82.9	65.8	59.0	77.2	64.4	31.0	73.6	29.5	70.9	28.1	68.3	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.2	17.3	-45.8	86.2	5.6	3.2	-49.9	88.5	-53.2	90.4	-56.3	92.1	-60.4	94.4		
(5)			-45.7	70.0	5.3	3.3	-49.5	72.3	-52.5	74.3	-55.5	76.1	-59.3	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	-6.9	-5.3	7.3	24.1	40.9	53.4	58.6	56.4	47.6	36.4	21.9	4.7
	DBStd	25.44	14.52	14.05	15.32	11.79	9.78	8.59	6.20	6.27	7.50	7.67	9.73	13.89
	HDD50	8610	1763	1549	1324	776	313	61	5	12	131	428	844	1404
	HDD65	13383	2228	1969	1789	1226	747	359	214	277	523	888	1294	1869
	CDD50	744	0	0	0	0	32	162	273	210	61	6	0	0
	CDD65	41	0	0	0	0	1	11	17	10	2	0	0	0
	CDH74	508	0	0	0	0	38	159	206	93	12	0	0	0
	CDH80	90	0	0	0	0	5	36	36	12	1	0	0	0
	WSAvg	7.8	7.1	7.1	7.4	7.9	7.8	7.9	7.6	7.7	8.8	8.3	8.5	7.5
	PrecAvg	29.9	1.4	0.9	1.1	1.3	1.9	3.0	4.5	4.3	4.6	3.1	2.3	1.7
(15) Precipitation	PrecMax	37.8	2.2	2.3	1.9	2.7	4.2	5.6	10.0	8.1	9.0	5.5	3.9	2.9
	PrecMin	24.1	0.5	0.4	0.2	0.4	0.5	0.9	2.3	2.7	1.2	1.3	1.2	0.6
	PrecStd	3.5	0.4	0.4	0.5	0.7	0.9	1.1	1.9	1.3	1.8	0.9	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	36.5	44.7	56.2	79.5	85.1	84.1	82.1	75.0	63.1	45.7	36.6
		MCWB	33.3	33.3	38.5	43.9	57.7	66.2	63.8	64.4	63.5	55.6	43.7	35.0
	2%	DB	27.8	30.4	38.7	49.5	71.4	78.3	79.3	76.7	68.8	56.5	41.0	32.4
		MCWB	26.9	28.0	34.0	40.1	53.9	59.9	61.9	62.6	60.4	52.1	39.2	31.5
	5%	DB	21.5	25.5	35.2	44.8	64.7	74.0	75.5	72.2	64.6	52.2	37.2	28.4
		MCWB	20.5	23.6	31.9	37.5	50.9	58.0	60.8	60.8	57.7	48.4	35.5	27.4
	10%	DB	16.4	17.7	31.2	40.9	58.9	69.9	71.8	68.5	60.5	48.1	34.3	24.5
		MCWB	15.4	16.3	28.3	35.0	48.1	56.3	59.6	59.5	55.0	44.9	33.2	23.5
	0.4%	WB	33.0	33.4	38.9	44.8	60.2	67.6	69.1	68.2	65.4	57.0	44.2	35.2
		MCDB	33.6	35.7	43.0	53.4	71.0	80.3	77.3	74.8	72.4	61.8	45.1	36.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.7	28.2	34.9	40.7	56.1	64.0	65.2	65.0	62.2	53.0	39.8	31.6
		MCDB	27.6	30.1	37.7	47.6	67.7	73.1	72.8	71.5	67.6	55.7	40.8	32.3
	5%	WB	20.5	23.4	31.9	38.2	52.5	61.3	63.3	62.9	59.0	49.0	35.6	27.3
		MCDB	21.5	25.4	35.1	43.9	63.2	68.9	70.5	69.9	63.5	51.8	36.8	28.2
	10%	WB	15.4	16.2	28.2	35.6	48.9	58.5	61.5	60.8	55.3	45.0	33.2	23.5
		MCDB	16.4	17.6	31.4	40.2	58.1	66.3	68.3	66.8	59.6	47.5	34.4	24.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	21.5	23.2	19.7	20.1	21.7	18.9	17.1	14.9	10.9	10.6	14.9
		MCDBR	19.3	19.6	21.1	21.7	31.6	28.8	26.8	23.9	21.7	18.2	12.0	12.8
		MCWBR	18.9	17.9	17.3	14.5	17.2	14.1	12.6	12.2	14.3	14.1	11.3	12.4
		MCDBR	18.9	19.5	19.2	19.4	27.1	21.8	19.6	18.5	18.9	16.2	11.3	12.9
	5% WB	MCWBR	18.6	18.0	16.3	13.8	16.1	12.9	11.6	11.3	14.3	13.8	11.0	12.6
(40) Clear-Sky Solar Irradiance	taub		0.237	0.251	0.274	0.327	0.365	0.365	0.390	0.362	0.351	0.297	0.270	0.227
	taud		2.205	2.227	2.201	2.145	2.307	2.386	2.358	2.441	2.437	2.534	2.247	2.186
	Ebn at Noon		237	273	289	285	279	280	270	272	260	257	217	215
	Edh at Noon		19	27	34	42	38	36	36	31	28	20	19	16
(44) All-Sky Solar Radiation	RadAvg		319	693	1208	1676	1718	1741	1511	1220	851	467	204	186
	RadStd		28	50	93	130	122	151	131	86	82	36	30	28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.37	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page